A The problem: the same speech is transcribed in two ways, and the way depends on diagnosis

PROCESS-2: 400 participants (200 HC, 150 MCI, 50 dementia), three tasks; example below is illustrative

What the participant says



"cat … dog, lion"

with a 3-s silence after "cat"

Convention 1: pauses written

```
Pat: cat (3 seconds)
     dog lion
```

Convention 2: pauses never written

```
cat dog lion
```

Sessions transcribed with convention 1



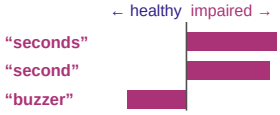
Missing annotations are not missing pauses: HC without them pause as much in the audio.

A text model can learn:

"seconds" → impaired

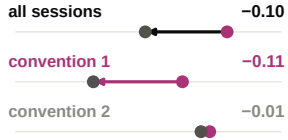
B The audit: is the gain due to the annotations or to the speech?

1 What does the model read?



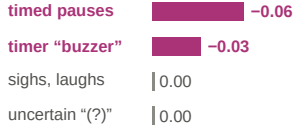
"seconds" is the top word in **3/3 tasks and 25/25 CV folds**

2 Remove the annotations



AUC lost only where annotations exist

3 Which annotations matter?



two categories carry the shortcut

4 Other checks

- demographics, channel **weak (AUC ≤ 0.56)**
- zero-shot language models **no prior use of annotations**
- silence fed alone to audio **≈ amount of pausing**
- cleaned transcripts **other habits remain**

C The control: a model that never reads a human annotation

Audio only



same input for everyone

Voice activity



pauses measured in audio

Speech recognition

cat dog lion

valid words per second

Healthy-control norms



z-scores by age and sex

Risk score



6 visible weights

Official test set, evaluated once: AUC 0.81 [0.70, 0.91]

same performance under both transcription conventions; all choices made on the training set